

Figure 9: Loadings (variables) in a 3D plot

```
Y=pdist(dataS,'euclidean');
Z=linkage(Y,'ward');
H=dendrogram(Z,0);
xlabel('Dendrogram (wcenter, euclidean, ward)');
```

For principal component analysis, *princomp* is used, whose parameters are (according to the help file):

- *facpr*, principal factors (eigenvectors of the correlation matrix);
- *comprinc*, principal components;
- *lambda*, eigenvalues of the correlation matrix;
- *tsquare*, Hotelling's statistic for each data point.

Then, some calculations are carried out to build a table (Figure 1) in which the first column represents an index, the second is the variance explained by each component, the third is the percentage of variance explained by each component and the fourth is the cumulative sum. Last, I can plot the percentage of variance, scores (*comprinc*, the objects) and loadings (*facpr*, the variables) using *plot* (Figures 5, 6 and 7) or *param3d1* (Figures 8 and 9).

```
clear;
sheets=readxls(uigetfile('*.xls'));
data=sheets(1).value;
dataS=wcenter(data,'r');
[facpr,comprinc,lambda,tsquare]=princomp(dataS);
n=length(lambda);
percentage=zeros(n,1);
runtot=zeros(n,1);
for i=1:n
    percentage(i)=lambda(i)/sum(lambda)*100;
end
runtot=cumsum(percentage);
s1=comprinc(:,1);
```

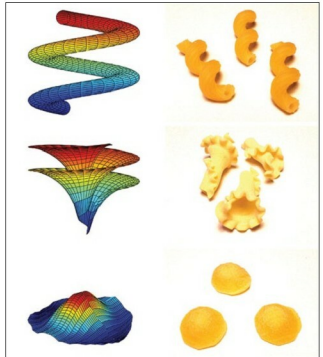


Figure 10: Pasta design, from top to bottom: corkscrews, lilies, little ears

```
s2=comprinc(:,2);
s3=comprinc(:,3);
s1=(1:1:length(s1));
l1=facpr(:,1);
l2=facpr(:,2);
l3=facpr(:,3);
l1=(1:1:length(l1));
printf(' N | VAR | PCT | SUM\n'..
+ '-----+-----+-----+-----\n');
for i=1:n
    printf(sprintf('%4.0f',i)+' ' |'..
+ sprintf('%10.4f',lambda(i))+' ' |'..
+ sprintf('%10.2f',percentage(i))+' ' |'..
+ sprintf('%10.2f',runtot(i))+' \n');
end
```

Pasta design

It is possible to think about pasta as a parametric surface. It's really very easy to plot. First, it's necessary to define the two parameters (*u* and *v*) and then the three functions (*x*, *y* and *z*). The pasta types under consideration here are: the simplified version of cavatappi (corkscrews); gigli (lilies), the simplified version based on the Dini's surface (from Ulisse Dini, 1845-1918); orecchiette (little ears), the simplified version based on the book 'Pasta by Design' by George L. Legendre (see References). You can refer to this book for many other examples and more complex equations. I can plot the surface using *plot3d1*. It's also necessary to set some other graphics parameters, like the colour map and the viewpoint (rotation angles). Last, I can compare each plot